Kavach AI — Your Invisible Shield

A passive safety system designed on the assumption that its own detector will sometimes be wrong.

Women's safety · Software Development track

TEAM THUNDERHAWKS

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CODE BUILD 1.0

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“We don't claim to distinguish an angry scream from a terrified one.
We built a system where it doesn't matter.”

Problem statement

Two problems, not one — and the second is the one nobody solves

The one action every safety app requires is the one action you cannot take.

WHY EVERY EXISTING APP FAILS

Panic buttons, the 112 India app, campus apps and wearable SOS all assume you can reach your phone and press something.

- **Restrained**
  Hands are held. The phone is unreachable.
- **Snatched**
  The phone is taken before you can act.
- **Frozen**
  A fear response stops deliberate action.

AND WHY "AI DETECTS A SCREAM" ALSO FAILS

- No classifier — ours, Google's, anyone's — can reliably tell an angry scream at a sibling from a terrified scream at an attacker.
- So it fires during arguments and loud family moments; guardians learn to ignore it.
- Auto-dialled police burn real emergency resources on noise. The tool destroys its own credibility.

So we stopped trying to build a perfect detector, and built a system that stays safe when the detector is wrong.

Proposed solution: a four-layer safety ladder

Detection never sends anything. Three independent layers surround it.

1 Context Engine

Decides how sceptical to be before detection matters. Place, time, route and which trusted devices are nearby set a dynamic threshold — capped at 85, so context can raise the bar but never close the door.

2 Detection

Dual-signal fusion of audio and motion, judged on the event's peak over an 800 ms window rather than the rising edge. Outputs a confidence score only; it can contact no one.

3 Sentinel Check

A dead-man's switch. The phone vibrates silently and asks you first, on a window that scales with context. Five ways to answer — tap pattern, unlock, safe word, shake, volume keys. Non-response becomes the trigger.

4 Guardian Verification

A soft “couldn't confirm she's okay” notice carrying the context that explains it, promoting to a full alert only if unresolved. A human confirms before any authority is contacted.

MANUAL OVERRIDE

A volume-button sequence, operable with the phone still in a pocket, bypasses all four layers — always. Context gates passive detection only; it can never block a deliberate call for help.

Workflow

1

LAYER 1

Context Engine

Decides how sceptical to be, before detection matters. Place, time and who is nearby set a dynamic threshold — **capped**, so it can never switch itself off.

2

LAYER 2

Detection

Audio and motion fused into one confidence score, judged on the event's peak. It outputs a number and **can contact no one**.

3

LAYER 3 · DEAD-MAN'S SWITCH

Sentinel Check

The phone asks **you** first, with a silent buzz. Five ways to answer — tap, unlock, safe word, shake, volume keys. **Non-response is the trigger.**

4

LAYER 4

Guardian Verification

A soft notice first, a full alert only if unresolved. **A human** sees location, audio and context — and confirms before any authority.

MANUAL TRIGGER A volume-button sequence bypasses all four layers — always. Context can never block a deliberate call for help.

Transit entry classification

Asks how the journey began: walking then driving is a commute — ride mode. A distress event then driving is a grab — abduction watch.

Repeat-suppression override

Three suppressed high-distress events inside the window open the sentinel check anyway. The pattern is the signal.

A trail, not a pin

In an abduction the valuable output is a live track police can follow, not a stale location from the moment of the grab.

Features, tech stack and vision

Every feature listed is implemented in the working prototype

KEY FEATURES

- **Five proof-of-life channels**
  Tap, unlock, safe word, shake, volume keys.
- **Duress pattern**
  Shows "Cancelled" while escalating silently.
- **Ride mode + crash detection**
  Motion discounted on rough roads; a crash still escalates.
- **Abduction watch**
  Guardians alerted at once, location streaming live.
- **Graded escalation + recall**
  Soft notice before full alert; recall corrects in seconds.
- **Journey and route watch**
  A missed ETA triggers a proof-of-life check.

TECH STACK

Client	Web Audio API, DeviceMotion, Geolocation — any phone browser
Audio	On-device audio event classifier (YAMNet class)
Motion	Accelerometer jerk and variance thresholding
Movement	ActivityRecognition / CMMotionActivity + GPS speed
Proximity	BLE RSSI scanning for registered contact devices
Alerting	Twilio or WhatsApp Business API, SMS fallback
Maps	OpenStreetMap embed; no API key required

VISION & LIMITS

Working prototype verified by 40 automated end-to-end tests · runs on any smartphone, no hardware cost · extends to elder care, child safety and lone-worker safety. Honest limit: a quiet abduction with no violence leaves passive detection nothing to detect — we would rather state where the system ends than pretend it doesn't.